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I will mainly use this as my project work.

Also a good chance to review the generalization of Linear Algebra to Module over PID ring!

Topics will include 1) Modules
2) Noetherian Ring / Module.
3) Module over PID (can perform decomposition)
4) Classification of Hom(V, V) (F is obviously PID R)

Mainly from Daniel's and my Note. Also some proofs refer to CLM 35: Advanced Linear Algebra.

Note some proof are just sketch version. (brief idea)

Also, may I will review category theory and Universal Property after them. Which is really elegant and convenient.

So LET US BEGIN!

Since write or edit through Latex will be time consuming. only handwrite version here

Topic 1. : Module | 1.1 Basic setting

Goal: from $\mathbb{F}$ to R.

Note here the R is integral domain (ID) where this R is commutative unital ring without any **zero divisor**

↓

$a \in R\setminus\{0\}$ s.t. $\exists b \in R\setminus\{0\}$ s.t. $a \cdot b = 0$

(Very important to avoid some bad thing.)

Example 1. $R = \mathbb{Z}_6$. $[2]\cdot[3] = 0_{\mathbb{Z}_6}$ Not a ID.

2. All field is ID. $a \cdot b = 0 \Rightarrow a^{-1} \cdot a \cdot b = 0 \Rightarrow b = 0$ (if $a \in \mathbb{F}\setminus\{0\}$)

3. if R is ID. $R[x]$ is also ID

Def) (R-Module) Given $(R, +, \cdot)$ as ID, M is a R-module, if

1) $(M, +)$ is abelian grmp ← "+"

2) Given $r, s \in R$, $u, v \in M$:

$$\begin{cases} r(u+v) = ru + rv \\ (r+s)u = ru + su \\ (r\cdot s)u = r\cdot(s\cdot u) \\ 1\cdot u = u \end{cases}$$

← "scalar ·"

Note def of M is same with V. ("U over R")

Example 1. $R = \mathbb{F}$, v.space are $\mathbb{F}$-Module

2. $R = \mathbb{Z}$, $M = \mathbb{Z}_n$.

$$\begin{cases} [a] + [b] := [a+b] \\ r\cdot[a] := [ra] \end{cases} \in \mathbb{Z}_n$$

3. $V = R^n$

$$\begin{cases} (r_1, \cdots, r_n) + (s_1, \cdots, s_n) := (r_1+s_1, \cdots, r_n+s_n) \\ a(r_1, \cdots, r_n) := (ar_1, \cdots, ar_n) \end{cases} \in R^n$$

Note in 2., $n \cdot [a] = 0$ so the "linearly independent" will be a great problem when generalize the whole theory on Modules!

4. An Important Example ⇝ related to $m_T(x)$ / $\chi_T(x)$ in 3.4.

Note $\mathcal{L}(V,V)$ is a V. space of $\mathbb{F}$, Consider $R(T)$: where $(T \in \mathcal{L}(V,V))$

$$R(T) := \{ p(T) := a_n T^n + \cdots + a_1 T + a_0 I \in \mathcal{L}(V,V) \mid a_i \in \mathbb{F}\ \forall i \}$$

(somehow it can be denoted as $\mathbb{F}[T]$)

Prop: $R(T)$ is a commutative unital subring of $\mathcal{L}(V,V)$

pf: NTS $\begin{cases} \alpha\, p(T) + \beta\, q(T) \in R(T) \\ p(T) \circ q(T) \in R(T) \end{cases}$

1. first one is trivial.
2. for second. define $p(T) \circ q(T) := (p \cdot q)(T)$ where $(p \cdot q)(x) := p(x) \cdot q(x)$

for $(p(T) \circ q(T))(v) = (p \cdot q)(T)(v) \in V$

Also $(p(T) \circ q(T))|_0 = (p \cdot q)(T) = (q \cdot p)(T) = q(T) \circ p(T)$

for i.d. denote as id. or 1.

□

Then one can define $T: V \to V$ (linear map.)

$M = V$ is a module over $\mathbb{F}[x]$ where $f(x) \cdot v := f(T)(v)$

Prop: for T above, always exist $p(x) \in \mathbb{F}[x]$ s.t. $p(T) = 0_{L(V,V)}$

p.f: $\dim(\mathcal{L}(V,V)) = n^2$

$\{I, T, \cdots, T^{n^2}\}$ is linearly dependent

$\Rightarrow a_0 + a_1 T + \cdots + a_{n^2} T^{n^2} = 0$ where some $a_0, \cdots, a_{n^2}$ all non-zero. ∎

1.2 Submodule & Ideal

Def (Submodule) Omitted (Similar with subspace.)

Def (Ideal) R is ID. $M = R$ be R-module. $I \triangleleft M (= R)$ submodules are called ideals of R in other word.

$i+j \in I$, $r \cdot i \in I$ for $i, j \in I$, $r \in R$

Not a subring.

Example $R = M = \mathbb{Z}$.

$I_1 = \{2k \mid k \in \mathbb{Z}\} = \langle\langle 2 \rangle\rangle = 2\mathbb{Z} \triangleleft \mathbb{Z}$.

$I_2 = \langle\langle 6, 9 \rangle\rangle = 3\mathbb{Z} \triangleleft \mathbb{Z} = \langle\langle 3 \rangle\rangle$

we gonna discuss it soon

1.3 Spanning Set & Linearly independence

Def (Spanning Set) M is R-Module. and $B \subseteq M$ is a subset

$\langle\langle B \rangle\rangle := \{r_1 b_1 + \cdots + r_n b_n \mid b_i \in B, r_i \in R\}$

if $|B| < \infty$: finitely generated / f.g.

if $|B| = \{b\}$ $\langle\langle b \rangle\rangle$ is 1-generated or cyclic submodule.

Example 1. $R = M = \mathbb{Z}$, $M = \langle\langle 1 \rangle\rangle$

2. $R = \mathbb{R}$, $M = R^n$ $M = \langle\langle e_1, \cdots, e_n \rangle\rangle$

3. $R = \mathbb{Z}[x]$ $M = R$ so $M = \langle\langle 1 \rangle\rangle$

but for $M' := \{p(x) \mid p(x)$ has a even costant term$\}$

$M' = \langle\langle 2, x \rangle\rangle$

Notice M' cannot reduce to 1 generated

4. Moreover, $R = \mathbb{Z}[x_1, x_2, \cdots]$ $M = R$ is 1-generated

$M' := \{p(x_1, x_2, \cdots) \mid p(0, \cdots, 0) = 0\}$

$= \langle\langle \{x_1, x_2, \cdots\} \rangle\rangle$

↳ ∞-generated

So a Fun fact, $M' \subseteq M$ "basis" of M' > "basis" of M

So here we need to introduce PID to avoid this situation.

Def (PID)

1. $I \trianglelefteq M (= R)$ is principal if $I = \langle\langle a \rangle\rangle$ is 1-generated
2. ID is called PID if all $I \trianglelefteq M (= R)$ is 1-generated.

Example 1. $R = \mathbb{Z}$ is PID probably the most important property we need (very strong)

$I \trianglelefteq R = n\mathbb{Z}$ $(n \in \mathbb{N})$

2. $R = \mathbb{F}[x]$ is PID

⇒ Note $I = \{0\}$ or if $I \neq \{0\}$ $f(x) = g(x)\,q(x) + r(x)$ (0 / smallest deg)

$I = \langle g(x) \rangle$?

3. $R = \mathbb{Z}[x]$ is not PID

Def (linear indept.) subset $S \subseteq M$ is linear indept. if any $\{s_1, \dots, s_n\} \subseteq S$.

$0 = r_1 s_1 + \dots + r_n s_n \iff 0 = r_1 = \dots = r_n$

Remark: It is different from v. space.

1. $R = \mathbb{Z}$ $M = \mathbb{Z}_n$. $[1]_{\mathbb{Z}}$ is linear depr
   ($n \cdot [1] = 0$) (V is idepr in v.space)
2. $\{v_1, v_2\}$ are depr. $v_2 = -k v_1$ ($k \in F$) in v.space

But $R = \mathbb{Z} = M$ $\{2, 3\} \Rightarrow$ $2 \cdot (-3) + 3 \cdot 2 = 0$

We gonna discuss real basis of M later when R is free Module. & None we keep to torsion and Ann
(torsion → kills r; Ann → kills m)

1.4 | Torsion & Annihilators

Def M be R-Module.

$$M_{tor} = \{ m \in M \mid r \cdot m = 0 \text{ for } r \in R \setminus \{0\} \}$$

- $M_{tor} = \{0\}$ → M is torsion-free ⇐ Good property for a basis.
- $M_{tor} = M$ → M is torsion module. ⇐ Bad

Prop: R is ID. $M_{tor} \subseteq M$ is submodule

pf: $\exists r_1, r_2 \neq 0$ s.t. $r_1 t_1 = r_2 t_2 = 0$ ($t_1, t_2 \in M_{tor}$)

$r_1 r_2 (p t_1 + q t_2) = 0$ (with $p t_1 + q t_2 \in M_{tor}$) ☐

"m be killed by some $r \in R$"

Def (Annihilator)

$\mathrm{Ann}(m) := \{ r \in R \mid r \cdot m = 0 \}$

$\mathrm{Ann}(M) := \{ r \in R \mid r \cdot m = 0 \ \forall m \in M \}$

Note. $\mathrm{Ann}(m) \lhd R$ is an ideal.

$r_1, r_2 \in \mathrm{Ann}(m)$ $\Rightarrow$ ✓ Same hold for $\mathrm{Ann}(M) \subseteq \mathrm{Ann}(m)$

$(r_1 + r_2)\, m = 0 \quad (r \cdot a)\, m = 0$

Example Let $R = F[x]$ $M = F[x]$-Module

if $f(x) \in \mathrm{Ann}(V)$ $m_T(x) \mid f(x)$

$\Rightarrow f(x) \in \langle\langle m_T(x) \rangle\rangle$

$\Rightarrow \mathrm{Ann}(V) = \langle\langle m_T(x) \rangle\rangle$

Now we have finish all set up, discuss of basis (most important of v space (free))

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6 Basis & free Module

Def M is free module if $\exists B$ which is spanning M linearly independently

Example: 1) $R = \mathbb{Z}$ $M = \mathbb{Z}_n$ $\Rightarrow$ Not torsion free

2) $M = R^n$ is free. with $\{e_1 \cdots e_n\}$

3) Let $R = \mathbb{Z}[x] = M$ is free by above example.

But $\langle\langle 2, x \rangle\rangle \lhd M = R$ is not free.

Note if M is free. B have same property as v space

Also, we can pf. all B of M have same cardinality, we never prof it.

Def $N \subseteq M$ be sub-M. $M/N := \{ m+N \mid m \in M \}$

for well defines, it is easy to check

Example : $M/N = \mathbb{Z}/n\mathbb{Z} = \{ a+n\mathbb{Z} \mid a \in \mathbb{Z} \}$

$= \{ 0+n\mathbb{Z}, \dots, (n-1)+n\mathbb{Z} \}$

and $n \cdot \bar{a} = \overline{n \cdot a} = \bar{0}$ for all $a \in M$. no basis

Theorem : M be R-Module. all bases of M have same cardinality

Def/prop : $I \lhd R$. if N either $= I$ or R, then $\mathbb{F} = (R/I, +, \cdot)$ is field

(needed)

Let $I \lhd R$ be maximal ideal. consider $IM := \{ i_1 m_1 + \cdots + i_k m_k \mid i_j \in I, m_k \in M, k \in \mathbb{N} \}$

$M/IM := \{ m + IM \mid m \in M \}$

Then R/I - M/IM is v-space where $(r+I)(m+IM) := (rm+IM)$

So we make following claim

if $B = \{ b_s \mid s \in S \}$. $C = \{ c_j \mid c_j \in J \}$ is L-bases of M.

then $B' = \{ b_s + IM \mid s \in S \}$, ... is L-bases of M/IM

if claim hold. it's obvious we are done.

proof of claim :

① Span : any $m \in M$

$m + IM = (r_1 b_1 + \cdots + r_n b_n) + IM$

$= (r_1+I)(b_1+IM) + \cdots + (r_n+I)(b_n+IM)$

② LI : let $(r_1+I)(b_1+IM) + \cdots + (r_n+I)(b_n+IM) = 0$

$\Rightarrow r_1 b_1 + \cdots + r_n b_n \in IM$

$= l_1 m_1 + \cdots + l_k m_k \quad (l \in I)$

for each $m_i = s_{i,1} b_1 + \cdots + s_{i,n} b_n$

So $r_1 b_1 + \cdots r_n b_n = (l_1 s_{1,1} + \cdots + l_n s_{n,1}) b_1 + \cdots$

$r_p = (l_1 s_{1,p} + \cdots + l_n s_{n,p})$

so each $r_p \in I$ $(r_p + I) = 0$ ☑

Now we nearly finished all theory related

Def (Homomorphism) M, N both R-Module $\phi: M \to N$ is an R-homomorphism if $\phi(r_1 m_1 + r_2 m_2) = r_1 \phi(m_1) + r_2 \phi(m_2)$

Moreover if ϕ is bijective, ϕ is an isomorphism over R

Denote as $Hom_R(M, N)$

Similar nearly all property is same as $(\mathbb{R}^n, +)$

I will omit this with matrix representation ... (All trivial)

Due to implement V over $\mathbb{F}$ to V over R more precise condition are needed

$$\begin{cases} \text{noether} \\ \text{f.g.} \\ \text{PID} \end{cases} \Rightarrow \begin{cases} M = M_{free} \oplus M_{tor} \\ \text{Primary decomposition} \\ \text{finite decomposition on primary module} \end{cases}$$

Due to the lack. I will only write our theorem direct and use it on $\mathbb{F}[x]$-Module and f.g. Abelian Group.

Theorem: Let R be PID, and M is f.g then $M = M_{free} \oplus M_{tor}$

Theorem: Let R be PID. $M = \langle\langle v \rangle\rangle$ is cyclic with $Ann(M) = Ann(v) = \langle a \rangle$

then 1) $\langle\langle v \rangle\rangle \cong R/\langle a \rangle$

2) $\forall \beta \in R \setminus \{0\}$, $Ann(\beta v) = \langle \frac{a}{gcd(a,\beta)} \rangle$

3) $\langle\langle v \rangle\rangle = \langle\langle \beta v \rangle\rangle \iff gcd(a,\beta) = 1$

pf 1) Define $\pi: R \to \langle\langle v \rangle\rangle$

$: r \mapsto rv \Rightarrow Im(\pi) = M \quad Ker(\pi) = Ann(v) = \langle a \rangle$

$\Rightarrow R/\langle a \rangle \cong M$

2) $\forall r \in Ann(\beta v)$.

$r \cdot (\beta v) = 0 \iff (\beta \cdot r) \cdot v = 0$

$\iff \beta r \in \langle a \rangle$

$\iff \beta r = s \cdot a \quad (s \in R)$

$\iff a \mid \beta \cdot r$

$\iff \frac{a}{gcd(a,\beta)} \mid \frac{\beta}{gcd} \cdot v$

$\iff \frac{a}{gcd(a,\beta)} \mid v$

$\iff r \in \langle \frac{a}{gcd(a,\beta)} \rangle$

3) Note $\langle\langle \beta v \rangle\rangle \subseteq \langle\langle v \rangle\rangle$

"$\Rightarrow$": $\langle\langle \beta v \rangle\rangle = \langle\langle v \rangle\rangle$

$\langle \frac{a}{gcd} \rangle = \langle a \rangle \Rightarrow gcd(a,\beta) = 1$

"$\in$" $\exists x, y \in R$ s.t. $xa + y\beta = 1$

$\Rightarrow v = xav + y\beta v = y\beta v \in \langle\langle \beta v \rangle\rangle$ ☑

Further more, one have (if $a = p_1^{e_1} \cdots p_k^{e_k}$) all p_i prime

1) $M = \langle\langle v \rangle\rangle = \langle\langle u_1 \rangle\rangle \oplus \cdots \oplus \langle\langle u_k \rangle\rangle$

2) $Ann(u_i) = \langle p_i^{e_i} \rangle$ (so that $\langle\langle u_i \rangle\rangle \cong R/\langle p_i^{e_i} \rangle$)

3) $M \cong R/\langle p_1^{e_1} \cdots p_k^{e_k} \rangle \cong R/\langle p_1^{e_1} \rangle \oplus \cdots \oplus R/\langle p_k^{e_k} \rangle$ Chinese Remainder Theorem

Idea of proof: $\alpha_k = \beta_k v$, $c(\beta_k) = p_1^{e_1} \cdots \widehat{p_k^{e_k}} \cdots p_k^{e_k}$

2) is trivial

1) ① $v = (r_1\beta_1 + \cdots + r_k\beta_k)v = r_1\beta_1 v + \cdots + r_k\beta_k v$ ← Because

$\Rightarrow \langle v \rangle \subseteq \langle u_1 \rangle + \cdots + \langle u_k \rangle$

all $\langle u_i \rangle \subseteq \langle v \rangle$ so sum also

② Consider $\delta_1 u_1 + \cdots + \delta_k u_k = 0$

$\beta_i(\delta_1 u_1 + \cdots + \delta_k u_k) = \beta_i \delta_i u_i = 0$

$\delta_i \in \mathrm{Ann}(\beta_i u_i) = \left\langle \frac{p_i^{e_i}}{\gcd(p_i^{e_i}, \beta_i)} \right\rangle = \langle p_i^{e_i} \rangle = \mathrm{Ann}(u_i)$

$\Rightarrow \delta_i \in \mathrm{Ann}(u_i)$ ∎

Theorem (Primary Decomposition)

R is PID. M is torsion R-module with $\mathrm{Ann}(M) = \langle a \rangle$. Then

$M = M_1 \oplus \cdots \oplus M_k$ where $\mathrm{Ann}(M_i) = \langle p_i^{e_i} \rangle$ moreover this decomposition is unique up to isomorphism

Idea: $M_k = \beta_k M$

① $\forall m \in M$, $m = (r_1\beta_1 + \cdots r_k\beta_k)m$ ✓

② $m_1 + \cdots + m_k = 0$

$\beta_i m_1 + \cdots + \beta_i m_k = 0 \Rightarrow \beta_i m_i = 0$

③ $p_i^{e_i} \in \langle \xi \rangle \Rightarrow p_i^{e_i} = \xi v \Rightarrow \xi \mid p_i^{e_i} \Rightarrow \xi = p_i^{f_i}$

if $f_i < e_i$: $(p_1^{e_1} \cdots p_i^{f_i} \cdots p_k^{e_k})\, m_i = 0$ ※

↖ $\in \mathrm{Ann}(M)$

Furthermore we can decomposition more.

Def) $\mathrm{Ann}(M) = \langle p^e \rangle \Rightarrow M$ is primary for prime $p \in R$

Theorem: For such primary module $M = \langle\langle v_1 \rangle\rangle \oplus \cdots \oplus \langle\langle v_n \rangle\rangle$

where $\mathrm{Ann}(\langle\langle v_i \rangle\rangle) = \langle p^{e_i} \rangle$ $\Rightarrow$ Also unique up to isomorphism

$(e = e_1 \geq \cdots \geq e_n)$

proof: omitted (too long) (use property of ACC)

So Now for R is P.I.D. Any M f.g.

$$M = M_{tor} \oplus M_{free} \cong R^l \oplus \left\{ R/\langle p_1^{e_{1,1}} \rangle \oplus \cdots \oplus R/\langle p_1^{e_{1,a_1}} \rangle \oplus \cdots \oplus R/\langle p_k^{e_{k,1}} \rangle \oplus \cdots \oplus R/\langle p_k^{e_{k,a_k}} \rangle \right\}$$

Corollary: (Smith Normal form)

$$M \cong R^l \oplus R/\langle d_1 \rangle \oplus \cdots \oplus R/\langle d_m \rangle$$

where $d_2 \mid d_1,\ d_3 \mid d_2, \cdots$

Example (on f.g. abelian group) $R = \mathbb{Z} \Rightarrow G \cong \mathbb{Z}^{l} \oplus (\cdots)$

Finally Back to $F[x]$-Module of (V)

$F[x] = \langle 1 \rangle$, V is torsion

so $V = (V_{1,1} \oplus \cdots \oplus V_{1,a_1}) \oplus \cdots \oplus (V_{k,a_k} \oplus \cdots \oplus V_{k,a_k})$

$V_1 \oplus \cdots \oplus V_k$ $\Rightarrow$ each $V_{i,j}$ is Tmodule

$$\Rightarrow T_{B,B} = \begin{pmatrix} (T|_{V_{1,1}})_{B_{1,1}} & & \\ & \ddots & \\ & & (T|_{V_{k,a_k}})_{B_{k,a_k}} \end{pmatrix}$$

$\Rightarrow \det(xI - T_{B,B}) = \prod_i \prod_j \det(xI - (T|_{V_{i,j}})_{B_{i,j}})$

Note $p_i(x)^{e_{i,j}} = x^d + \beta_{d-1}x^{d-1} + \cdots + \beta_0$

define $\phi : V_{i,j} \xrightarrow{\cong} F[x]/\langle p_i(x)^{e_{i,j}} \rangle$

$\phi(w^j) = x^j + \langle p_i(x)^{e_{i,j}} \rangle$

$$\Rightarrow \phi(T(w^j)) = \begin{cases} \phi(w^{j+1}) & j < d-1 \\ \phi(-\beta_{d-1}w^{d-1} - \cdots - \beta_0) & j = d-1 \end{cases}$$

$$\Rightarrow (T|_{V_{i,j}})_{B_{i,j}} = \begin{pmatrix} 0 & & & -\beta_0 \\ 1 & \ddots & & \vdots \\ & \ddots & 0 & \\ & & 1 & -\beta_{d-1} \end{pmatrix}$$

Cayley Hamilton

by direct computation, $\det(xI - T_{B,B}) = \prod_i \prod_j p_i(x)^{e_{i,j}}$

also rational normal form are proved.

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for Jordan normal form $\Rightarrow m_T(x) = (x-a_1)^{e_1} \cdots (x-a_k)^{e_k}$

$CF = JF$

Define $\phi(w^j) = (x-a_i)^j + \langle (x-a_i)^{e_{i,i}} \rangle$ we are done !